FPE Data Model Schema v3.0

Finance-First F&B Projection Engine

Document Type: Unified Master Data Model Schema
Version: v3.0
Industry: Food & Beverage — Dark Kitchen / Pizza Format
Architecture: Finance-First, Driver-Based, Scenario-Driven
Core Tables: 62 core tables + 11 optional extension tables = 73 total
Schema Layers: 10 core layers (A–J) + 1 optional extension layer
Date: 2026
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1. Introduction & Design Philosophy

This document defines the complete data model schema for the Finance-First F&B Projection Engine (FPE). The engine is purpose-built for dark-kitchen food & beverage operations, starting with the pizza format, and is designed to support financial planning, scenario analysis, market expansion, and performance intelligence across a 6-to-18-month decision horizon.

1.1 Finance-First Design Principles

Driver-Based Modeling: All financial outputs are computed from explicit, auditable assumption inputs rather than from raw transaction data. Every metric traces back to a named driver.

Periodization Over Granularity: The planning unit is a Planning Period (typically one calendar month). Intra-day and order-level granularity are optional extensions, not core.

Scenario-Driven Versioning: Every assumption set, driver record, and projection is attached to a Scenario. Multiple scenarios co-exist and can be compared side-by-side.

Layered Architecture: The schema is organized in 10 logical layers: Planning Spine -> Reference Dimensions -> Driver Engines -> Projection Engine -> KPI/Intelligence -> Risk -> Simulation -> Portfolio.

Best Simplification Rule: Exclude any detail that does not materially affect a financial decision within the 6-to-18-month horizon. Operational detail lives in the Optional Extension layer.

Assumption Confidence: Every assumption record carries a confidence score and volatility flag, enabling uncertainty quantification across all downstream projections.

Multi-Tenancy & Governance: All tables include company_id for tenant isolation. Plan versions carry approval status, owner, and freeze timestamps for governance compliance.

1.2 Planning Grain Evolution

The planning grain evolves as the business matures:

Stage	Grain	Dimension Keys	Typical Period
Pre-Launch	Market x Kitchen Format x Month	market_id, kitchen_format, period_id	Monthly
Early Ramp	Market x Platform x Product Family x Month	market_id, platform_id, product_family_id, period_id	Monthly
Steady State	Kitchen x Platform x Product Family x Month	kitchen_id, platform_id, product_family_id, period_id	Monthly / Weekly
Portfolio	Company x Country x Cluster x Month	company_id, country_id, cluster_id, period_id	Monthly / Quarterly

1.3 Schema Conventions

Convention	Rule
Primary Keys	UUID via gen_random_uuid(). All PKs named <entity>_id.

Money Fields	DECIMAL(19,4) — e.g., revenue_amount, cost_amount. Currency stored in currency_code (ISO 4217).
Percentage Fields	DECIMAL(5,4) — e.g., 0.1500 = 15%. Range 0.0000 to 9.9999.
Ratio / Score Fields	DECIMAL(8,6) for high-precision ratios; DECIMAL(5,2) for scores 0-100.
Timestamps	TIMESTAMPTZ (UTC). Columns: created_at, updated_at. All periods use period_start_date / period_end_date (DATE).
Audit Trail	Every table: created_by UUID, updated_by UUID, created_at TIMESTAMPTZ, updated_at TIMESTAMPTZ.
Soft Deletes	is_deleted BOOLEAN DEFAULT FALSE; deleted_at TIMESTAMPTZ; deleted_by UUID.
Multi-Tenancy	company_id UUID NOT NULL on every table. Indexed.
JSONB Metadata	metadata JSONB for flexible extensibility without schema migrations.
Enums	Defined as PostgreSQL ENUM types. All enum names prefixed with fpe_. See Section 11 for full enum reference.
Foreign Keys	Named fk_<table>_<referenced_table>. ON DELETE RESTRICT for spine entities; ON DELETE CASCADE for driver/projection facts.
Indexes	Partial indexes on (company_id, is_deleted=FALSE) for all tenant queries. Composite indexes on planning grain keys.
Naming	snake_case for all identifiers. Tables are plural nouns. Junction tables named <table_a>_<table_b>.
Comments	COMMENT ON TABLE and COMMENT ON COLUMN for every entity and field.

2. Schema Architecture Overview

The schema is organized into 11 logical layers. The first 10 layers constitute the core schema (62 tables). Layer 11 is the Optional Extension layer (11 tables) for operational detail beyond the 6-18 month planning horizon.

Layer	Code	Name	Tables	Purpose
A	PSG	Planning Spine & Governance	8	Foundation entities: company, calendar, periods, scenarios, assumption sets, plan versions, governance events, audit log.
B	PMR	Product & Market Reference	8	Reference dimensions: markets, kitchens, platforms, product families, revenue streams, menu hierarchies.
C	CDE	Commercial Driver Engine	5	Demand, pricing, product mix, platform mix, and revenue projection drivers.
D	CTE	Cost Driver Engine	7	Unit cost, labor, headcount, OPEX, marketing, CAPEX, and working capital drivers.
E	FCS	Funding & Capital Strategy	5	Equity rounds, debt facilities, repayment schedules, capital deployment plans, treasury policies.
F	FPE	Financial Projection Engine	4	Projected P&L, cash flow statement, balance sheet, and revenue breakdown projections.
G	KPI	KPI & Performance Intelligence	4	KPI projections, scenario comparisons, driver explainability, and performance alerts.
H	RUF	Risk & Uncertainty Framework	5	Risk scenarios, risk registers, uncertainty bands, stress test definitions, risk trigger log.

I	SDS	Simulation & Decision Support	5	Simulation runs, sensitivity analysis, Monte Carlo outputs, breakeven analysis, decision memory.
J	PCO	Portfolio Construction	4	Portfolio scenarios, market portfolio allocations, cross-market synergies, capital allocation.
EXT	OEL	Optional Extension Layer	11	Operational detail deferred to Phase 6+: orders, recipes, variants, kitchen stations, intraday forecasts.

2.1 Cross-Layer Data Lineage

Data flows through the schema in a directed acyclic graph from assumptions to financial outputs:

From Layer	To Layer	Key Relationship
A — Planning Spine	C, D, E — Driver Engines	Scenario + Assumption Set + Planning Period contextualize all driver records
B — Reference Dimensions	C, D — Driver Engines	market_id, kitchen_id, platform_id, product_family_id scope all driver facts
C — Commercial Drivers	F — Financial Projections	Demand + Price + Mix drivers compute Revenue Projection Lines
D — Cost Drivers	F — Financial Projections	Unit Cost + Labor + OPEX + CAPEX drivers compute cost line items in P&L
E — Funding & Capital	F — Financial Projections	Equity, debt, and treasury policies compute cash flow and balance sheet
F — Financial Projections	G — KPI Intelligence	Projected P&L, cash flow, and balance sheet feed KPI computations
G — KPI Intelligence	H — Risk Framework	KPI thresholds define risk breach conditions
H — Risk Framework	I — Simulation	Risk scenarios parameterize stress tests and Monte Carlo runs
I — Simulation	J — Portfolio	Simulation outputs inform portfolio capital allocation decisions
A — Planning Spine	I — Simulation	Scenario versioning governs simulation parameterization

3. Layer A — Planning Spine & Governance

The Planning Spine is the foundation of the entire schema. Every driver record, projection fact, KPI value, and simulation output must reference a valid combination of (company_id, scenario_id, assumption_set_id, planning_period_id). This composite key — the Planning Grain Key — ensures all data is versioned, scenario-isolated, and periodized.

Tables: companies, planning_calendars, planning_periods, scenarios, assumption_sets, plan_versions, governance_events, planning_audit_log

Core Key: company_id + scenario_id + assumption_set_id + planning_period_id

3.1 companies

Root tenant entity. All other entities belong to a company.

Field	Type	Constraints	Description
company_id	UUID	PK, NOT NULL	Unique company identifier
company_name	VARCHAR(200)	NOT NULL	Legal entity name
company_code	VARCHAR(20)	UNIQUE NOT NULL	Short code (e.g., FPE-001)
industry_vertical	fpe_industry_vertical	NOT NULL	e.g., food_and_beverage
base_currency	CHAR(3)	NOT NULL DEFAULT 'AED'	ISO 4217 currency code
fiscal_year_start_month	SMALLINT	NOT NULL DEFAULT 1	1=Jan, 7=Jul, etc.
default_planning_horizon_months	SMALLINT	NOT NULL DEFAULT 18	Default forecast window
country_of_incorporation	CHAR(2)	NOT NULL	ISO 3166-1 alpha-2
status	fpe_company_status	NOT NULL DEFAULT 'active'	active, suspended, archived
metadata	JSONB	DEFAULT '{}'	Flexible extension fields
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL	Last update timestamp
created_by	UUID	NOT NULL	User who created this record
updated_by	UUID	NOT NULL	User who last updated this record
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	Soft delete flag
deleted_at	TIMESTAMPTZ		Soft delete timestamp

3.2 planning_calendars

Defines the time structure for planning. A company may have multiple calendars (e.g., financial year, tax year).

Field	Type	Constraints	Description

calendar_id	UUID	PK, NOT NULL	Calendar identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
calendar_name	VARCHAR(100)	NOT NULL	e.g., FY2025 Planning Calendar
calendar_type	fpe_calendar_type	NOT NULL	gregorian, fiscal, tax
start_date	DATE	NOT NULL	Calendar start date
end_date	DATE	NOT NULL	Calendar end date
default_period_type	fpe_period_type	NOT NULL DEFAULT 'monthly'	monthly, quarterly, annual
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	Whether calendar is in use
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

3.3 planning_periods

Individual planning time units within a calendar. Each period is the atomic time unit for all driver and projection facts.

Field	Type	Constraints	Description
period_id	UUID	PK, NOT NULL	Period identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
calendar_id	UUID	FK -> planning_calendars, NOT NULL	Parent calendar
period_label	VARCHAR(50)	NOT NULL	e.g., 2025-M01, 2025-Q1
period_type	fpe_period_type	NOT NULL	monthly, quarterly, annual, weekly
period_start_date	DATE	NOT NULL	Inclusive start date
period_end_date	DATE	NOT NULL	Inclusive end date
fiscal_year	SMALLINT	NOT NULL	Fiscal year number
fiscal_quarter	SMALLINT		1-4 (NULL for non-quarterly periods)
fiscal_month	SMALLINT		1-12 (NULL for non-monthly periods)
is_actuals_available	BOOLEAN	NOT NULL DEFAULT FALSE	True when actuals have been ingested
trading_days	SMALLINT	NOT NULL	Number of trading days in period
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

3.4 scenarios

Named planning scenarios. All driver and projection data is scenario-scoped, enabling parallel versions of the financial plan.

Field	Type	Constraints	Description
scenario_id	UUID	PK, NOT NULL	Scenario identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_name	VARCHAR(100)	NOT NULL	e.g., Base Case 2025, Bull Case, Stress Test Q3
scenario_code	VARCHAR(30)	UNIQUE NOT NULL	Short code, e.g., BASE-2025
scenario_type	fpe_scenario_type	NOT NULL	base, upside, downside, stress, budget, forecast, actuals
parent_scenario_id	UUID	FK -> scenarios, NULLABLE	For cloned/derived scenarios
clone_depth	SMALLINT	NOT NULL DEFAULT 0	0=original, 1=first clone, etc.
planning_horizon_months	SMALLINT	NOT NULL DEFAULT 18	How far forward this scenario projects
base_currency	CHAR(3)	NOT NULL	Scenario currency (may differ from company default)
status	fpe_scenario_status	NOT NULL DEFAULT 'draft'	draft, in_review, approved, published, archived, frozen
approval_status	fpe_approval_status	NOT NULL DEFAULT 'pending'	pending, approved, rejected
approved_by	UUID		User who approved
approved_at	TIMESTAMPTZ		Approval timestamp
frozen_at	TIMESTAMPTZ		When scenario was frozen (no further edits)
description	TEXT		Narrative description of scenario assumptions
tags	TEXT[]	DEFAULT '{}'	Searchable tags
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

3.5 assumption_sets

A versioned collection of assumptions that parameterizes a scenario. A scenario may have multiple assumption sets over time, enabling budget-vs-forecast comparisons.

Field	Type	Constraints	Description
assumption_set_id	UUID	PK, NOT NULL	Assumption set identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Parent scenario
set_name	VARCHAR(100)	NOT NULL	e.g., Initial Budget, Mid-Year Reforecast
set_version	VARCHAR(20)	NOT NULL DEFAULT 'v1.0'	Semantic version
set_type	fpe_assumption_set_type	NOT NULL	budget, forecast, actuals_aligned, stress
overall_confidence_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Aggregate confidence 0-100
volatility_flag	fpe_volatility_level	NOT NULL DEFAULT 'medium'	low, medium, high, very_high
owner_user_id	UUID	NOT NULL	Primary owner / analyst
reviewer_user_id	UUID		Assigned reviewer
approval_status	fpe_approval_status	NOT NULL DEFAULT 'pending'	pending, approved, rejected
approved_by	UUID		
approved_at	TIMESTAMPTZ		
is_frozen	BOOLEAN	NOT NULL DEFAULT FALSE	Frozen sets cannot be edited
frozen_at	TIMESTAMPTZ		
valid_from_period_id	UUID	FK -> planning_periods	First period this set applies to
valid_to_period_id	UUID	FK -> planning_periods, NULLABLE	Last period (NULL = open-ended)
notes	TEXT		Analyst notes on key assumption changes
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

3.6 plan_versions

Immutable snapshots of a plan at a point in time. Used for audit, regulatory submissions, and board presentations.

Field	Type	Constraints	Description

version_id	UUID	PK, NOT NULL	Version snapshot identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario this version belongs to
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set snapshotted
version_label	VARCHAR(50)	NOT NULL	e.g., Board Pack v3, Investor Data Room
version_number	VARCHAR(20)	NOT NULL	Semantic version string
snapshot_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When snapshot was taken
snapshot_by	UUID	NOT NULL	User who created snapshot
snapshot_reason	TEXT		Why this snapshot was created
is_published	BOOLEAN	NOT NULL DEFAULT FALSE	Published versions are locked
published_at	TIMESTAMPTZ		
published_by	UUID		
storage_path	TEXT		Optional: path to exported PDF/XLSX
checksum	VARCHAR(64)		SHA-256 of snapshot JSON for integrity
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

3.7 governance_events

Records all governance actions: approvals, rejections, escalations, lock/unlock events on scenarios, assumption sets, and plan versions.

Field	Type	Constraints	Description
event_id	UUID	PK, NOT NULL	Event identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
entity_type	fpe_governance_entity_type	NOT NULL	scenario, assumption_set, plan_version, driver_record
entity_id	UUID	NOT NULL	ID of the entity this event pertains to
event_type	fpe_governance_event_type	NOT NULL	submitted, approved, rejected, frozen, published, escalated, recalled
actor_user_id	UUID	NOT NULL	User performing the action

delegated_to_user_id	UUID		If approval was delegated
event_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When event occurred
comments	TEXT		Governance notes or rejection reason
previous_status	VARCHAR(50)		Status before this event
new_status	VARCHAR(50)		Status after this event
metadata	JSONB	DEFAULT '{}'	Any additional context
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

3.8 planning_audit_log

Immutable append-only log of all changes to planning data. Supports regulatory compliance, debugging, and change history.

Field	Type	Constraints	Description
log_id	UUID	PK, NOT NULL	Log entry identifier
company_id	UUID	NOT NULL	Tenant reference
table_name	VARCHAR(100)	NOT NULL	Table where change occurred
record_id	UUID	NOT NULL	PK of the changed record
operation	fpe_audit_operation	NOT NULL	INSERT, UPDATE, DELETE
changed_by	UUID	NOT NULL	User who made the change
changed_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	Timestamp of change
old_values	JSONB		Previous field values (for UPDATE/DELETE)
new_values	JSONB		New field values (for INSERT/UPDATE)
session_id	UUID		Application session identifier
ip_address	INET		Client IP address
user_agent	TEXT		Application user agent string
scenario_id	UUID		Scenario context if applicable
assumption_set_id	UUID		Assumption set context if applicable

4. Layer B — Product & Market Reference

Reference dimensions define the geographic, operational, and commercial scope of the planning model. These are relatively stable master data entities that contextualize all driver and projection facts.

Tables: markets, kitchens, platforms, product_families, revenue_streams, menu_hierarchies, market_attractiveness_scores, kitchen_formats

4.1 markets

Geographic market units at any level of the hierarchy (Country, Cluster/Region, Macro/City, Micro/Area).

Field	Type	Constraints	Description
market_id	UUID	PK, NOT NULL	Market identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
market_name	VARCHAR(200)	NOT NULL	Human-readable name
market_code	VARCHAR(30)	UNIQUE NOT NULL	Short code, e.g., AE-DXB-JLT-N
market_level	fpe_market_level	NOT NULL	country, cluster, macro, micro
parent_market_id	UUID	FK -> markets, NULLABLE	Parent in geographic hierarchy
country_code	CHAR(2)	NOT NULL	ISO 3166-1 alpha-2
currency_code	CHAR(3)	NOT NULL	Local currency
timezone	VARCHAR(50)	NOT NULL	IANA timezone string
latitude	DECIMAL(9,6)		Centroid latitude
longitude	DECIMAL(9,6)		Centroid longitude
population_estimate	BIGINT		Latest population estimate
target_demographic_pct	DECIMAL(5,4)		Estimated addressable demographic share
competition_intensity_score	DECIMAL(5,2)		0-100 competition intensity
price_power_index	DECIMAL(5,2)		Relative price power vs. company average
labor_tightness_score	DECIMAL(5,2)		0-100; 100=very tight labor market
real_estate_cost_index	DECIMAL(7,2)		Indexed cost vs. base market
platform_penetration_pct	DECIMAL(5,4)		Share of target demographic on delivery platforms
regulatory_complexity_score	DECIMAL(5,2)		0-100
status	fpe_market_status	NOT NULL DEFAULT 'prospective'	prospective, approved, active, paused, exited
launch_target_date	DATE		Planned launch date

actual_launch_date	DATE		Actual launch date
metadata	JSONB	DEFAULT '{}'	Flexible fields
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.2 kitchens

Physical or virtual kitchen units. Each kitchen is the operational unit for capacity, cost, and financial projections.

Field	Type	Constraints	Description
kitchen_id	UUID	PK, NOT NULL	Kitchen identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
market_id	UUID	FK -> markets, NOT NULL	Micro-level market this kitchen serves
kitchen_name	VARCHAR(200)	NOT NULL	e.g., JLT North Dark Kitchen
kitchen_code	VARCHAR(30)	UNIQUE NOT NULL	e.g., DXB-JLT-001
kitchen_format_id	UUID	FK -> kitchen_formats, NOT NULL	Format specification
status	fpe_kitchen_status	NOT NULL DEFAULT 'planned'	planned, under_fit_out, operational, paused, closed
planned_open_date	DATE		Planned operational start date
actual_open_date	DATE		Actual operational start date
floor_area_sqm	DECIMAL(8,2)		Total floor area in square meters
kitchen_area_sqm	DECIMAL(8,2)		Net kitchen production area
monthly_rent_amount	DECIMAL(19,4)		Monthly rent in base currency
rent_currency	CHAR(3)		Currency of rent agreement
lease_start_date	DATE		
lease_end_date	DATE		
max_theoretical_orders_per_day	INT		Theoretical maximum daily orders
practical_capacity_utilization_pct	DECIMAL(5,4)		Practical capacity as pct of theoretical
latitude	DECIMAL(9,6)		Kitchen GPS latitude
longitude	DECIMAL(9,6)		Kitchen GPS longitude

metadata	JSONB	DEFAULT '{}'	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.3 platforms

Delivery platforms and owned channels through which products are sold.

Field	Type	Constraints	Description
platform_id	UUID	PK, NOT NULL	Platform identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
platform_name	VARCHAR(100)	NOT NULL	e.g., Talabat, Deliveroo, Own App
platform_code	VARCHAR(30)	UNIQUE NOT NULL	e.g., TALABAT-AE
platform_type	fpe_platform_type	NOT NULL	third_party_aggregator, owned_app, owned_web, direct_phone, b2b
commission_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Platform commission as decimal
payment_terms_days	SMALLINT	NOT NULL DEFAULT 14	Days from delivery to cash receipt
average_discount_pct	DECIMAL(5,4)	DEFAULT 0	Average platform-funded discount
self_funded_discount_pct	DECIMAL(5,4)	DEFAULT 0	Average brand-funded discount
base_delivery_fee	DECIMAL(10,4)	DEFAULT 0	Base delivery fee charged to customer
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
contract_start_date	DATE		
contract_end_date	DATE		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.4 product_families

High-level product groupings used for planning-grain aggregation. Examples: Pizza, Sides, Beverages, Desserts.

Field	Type	Constraints	Description

product_family_id	UUID	PK, NOT NULL	Product family identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
family_name	VARCHAR(100)	NOT NULL	e.g., Pizza, Sides, Beverages
family_code	VARCHAR(30)	UNIQUE NOT NULL	e.g., FAM-PIZZA
family_type	fpe_product_family_type	NOT NULL	core, complementary, premium, limited_time, b2b
average_food_cost_pct	DECIMAL(5,4)	NOT NULL	Average COGS as % of net revenue
average_prep_time_minutes	DECIMAL(5,2)		Average kitchen prep time
average_order_contribution_pct	DECIMAL(5,4)		Typical % of orders containing this family
vat_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.05	VAT rate applicable (5% UAE default)
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
launch_date	DATE		When family was/will be available
metadata	JSONB	DEFAULT '{}'	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.5 revenue_streams

Defines revenue generation models (e.g., delivery orders, catering, B2B, subscriptions).

Field	Type	Constraints	Description
stream_id	UUID	PK, NOT NULL	Revenue stream identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
stream_name	VARCHAR(100)	NOT NULL	e.g., Platform Delivery, Catering, B2B
stream_code	VARCHAR(30)	UNIQUE NOT NULL	e.g., STR-PLATFORM-DELIVERY
stream_type	fpe_revenue_stream_type	NOT NULL	platform_delivery, own_channel, catering, b2b, subscription, other
recognition_basis	fpe_revenue_recognition	NOT NULL	on_delivery, on_payment, ratable, milestone
typical_aov_amount	DECIMAL(19,4)		Typical average order value
typical_margin_pct	DECIMAL(5,4)		Typical contribution margin

is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.6 menu_hierarchies

Captures the menu structure for planning purposes: Collection -> Category -> Product Family. Detailed SKUs are in the Optional Extension layer.

Field	Type	Constraints	Description
menu_node_id	UUID	PK, NOT NULL	Menu node identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
node_name	VARCHAR(200)	NOT NULL	Node display name
node_type	fpe_menu_node_type	NOT NULL	menu, collection, category, product_family
parent_node_id	UUID	FK -> menu_hierarchies, NULLABLE	Parent node
product_family_id	UUID	FK -> product_families, NULLABLE	Links leaf nodes to product families
display_order	SMALLINT	NOT NULL DEFAULT 0	Sort order within parent
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.7 market_attractiveness_scores

Stores attractiveness scoring inputs and outputs for each market, supporting the expansion decision framework.

Field	Type	Constraints	Description
score_id	UUID	PK, NOT NULL	Score record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
market_id	UUID	FK -> markets, NOT NULL	Market being scored
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
score_date	DATE	NOT NULL	Date of scoring exercise
demand_potential_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Demand attractiveness 0-100

competition_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Lower score = more competitive
operational_feasibility_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Ease of operations
financial_return_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Projected return potential
risk_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Lower = higher risk
strategic_fit_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Alignment with portfolio strategy
composite_score	DECIMAL(5,2)	GENERATED	Weighted composite of sub-scores
scoring_weights	JSONB	NOT NULL	Weight for each dimension
recommendation	fpe_expansion_recommendation		enter, monitor, defer, avoid
scored_by	UUID	NOT NULL	User or system that scored
notes	TEXT		Scoring narrative
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

4.8 kitchen_formats

Defines kitchen format templates specifying standard capacity, cost, and capex profiles.

Field	Type	Constraints	Description
format_id	UUID	PK, NOT NULL	Format identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
format_name	VARCHAR(100)	NOT NULL	e.g., Standard Dark Kitchen, Ghost Kitchen Hub
format_code	VARCHAR(30)	UNIQUE NOT NULL	e.g., FMT-SDK-001
min_area_sqm	DECIMAL(8,2)	NOT NULL	Minimum viable floor area
max_area_sqm	DECIMAL(8,2)	NOT NULL	Maximum standard area
standard_capex_amount	DECIMAL(19,4)	NOT NULL	Standard fit-out + equipment capex
standard_monthly_opex_amount	DECIMAL(19,4)	NOT NULL	Fixed monthly OPEX baseline
theoretical_max_orders_per_day	INT	NOT NULL	At standard configuration
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	

is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

5. Layer C — Commercial Driver Engine

The Commercial Driver Engine captures all assumptions and calculations that drive the revenue side of the model. For each planning grain key (company, scenario, assumption set, period, market/kitchen, platform, product family), the engine combines demand volume, pricing, and mix assumptions to produce revenue projection line items.

Tables: demand_drivers, price_plans, mix_plans, seasonality_profiles, revenue_projection_lines

Revenue Formula: Net Revenue = Orders x AOV x (1 - commission_rate) x (1 - discount_rate) x (1 - cancellation_rate)

5.1 demand_drivers

Core demand assumptions per planning grain. Drives the orders volume that flows into all downstream calculations.

Field	Type	Constraints	Description
driver_id	UUID	PK, NOT NULL	Driver record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope (NULL=market-level)
platform_id	UUID	FK -> platforms, NOT NULL	Platform scope
product_family_id	UUID	FK -> product_families, NOT NULL	Product family scope
baseline_orders_per_day	DECIMAL(12,4)	NOT NULL	Base daily orders at period start
organic_growth_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Period-over-period organic growth
marketing_uplift_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Demand uplift from marketing spend
penetration_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Market penetration (active customers / total addressable)
market_ramp_factor	DECIMAL(5,4)	NOT NULL DEFAULT 1.0	Ramp factor for new markets (0-1)
seasonality_index	DECIMAL(5,4)	NOT NULL DEFAULT 1.0	Seasonal multiplier for this period
price_elasticity_coefficient	DECIMAL(8,6)	NOT NULL DEFAULT -1.0	Own-price elasticity
platform_share_pct	DECIMAL(5,4)	NOT NULL	Platform's share of total market orders

projected_orders_per_day	DECIMAL(12,4)	GENERATED	Computed daily orders (baseline x growth x uplift x seasonality x ramp)
projected_orders_total	DECIMAL(14,4)	GENERATED	Total orders for period (orders_per_day x trading_days)
confidence_score	DECIMAL(5,2)	CHECK (0<=x<=100)	Driver confidence 0-100
volatility_flag	fpe_volatility_level	NOT NULL DEFAULT 'medium'	low, medium, high, very_high
data_source	VARCHAR(200)		Source of demand assumption
notes	TEXT		Analyst notes
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

5.2 price_plans

Pricing assumptions per planning grain. Defines AOV, discount structures, and pricing strategy inputs.

Field	Type	Constraints	Description
price_plan_id	UUID	PK, NOT NULL	Price plan identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
platform_id	UUID	FK -> platforms, NOT NULL	Platform scope
product_family_id	UUID	FK -> product_families, NOT NULL	Product family scope
gross_aov_amount	DECIMAL(19,4)	NOT NULL	Gross average order value before discounts
platform_discount_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Platform-funded discount rate
brand_discount_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Brand-funded discount rate
net_aov_amount	DECIMAL(19,4)	GENERATED	gross_aov x (1 - platform_disc - brand_disc)

vat_inclusive	BOOLEAN	NOT NULL DEFAULT TRUE	Whether AOV includes VAT
vat_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.05	Applicable VAT rate
cancellation_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.02	Expected order cancellation rate
refund_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.01	Expected refund rate
price_change_reason	TEXT		Rationale for pricing vs. prior period
confidence_score	DECIMAL(5,2)		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

5.3 mix_plans

Defines the product family and platform mix assumptions — what proportion of orders come from each product family and each platform.

Field	Type	Constraints	Description
mix_plan_id	UUID	PK, NOT NULL	Mix plan identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
product_family_id	UUID	FK -> product_families, NOT NULL	Product family
platform_id	UUID	FK -> platforms, NOT NULL	Platform
order_mix_pct	DECIMAL(5,4)	NOT NULL	% of total orders from this family x platform combination
revenue_mix_pct	DECIMAL(5,4)	NOT NULL	% of total revenue from this combination
cogs_mix_pct	DECIMAL(5,4)	NOT NULL	% of total COGS from this combination
basket_attach_rate_pct	DECIMAL(5,4)	DEFAULT 0	Likelihood another family is attached to same order

mix_change_driver	TEXT		Narrative on what drives mix change vs prior period
confidence_score	DECIMAL(5,2)		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
updated_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

5.4 seasonality_profiles

Seasonal adjustment factors applied to demand drivers. Profiles can be shared across markets or defined per market.

Field	Type	Constraints	Description
profile_id	UUID	PK, NOT NULL	Profile identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
profile_name	VARCHAR(100)	NOT NULL	e.g., UAE Standard, Ramadan Profile
market_id	UUID	FK -> markets, NULLABLE	NULL = global default
product_family_id	UUID	FK -> product_families, NULLABLE	NULL = all families
period_month	SMALLINT	CHECK (1<=x<=12)	Calendar month
seasonality_index	DECIMAL(5,4)	NOT NULL DEFAULT 1.0	1.0 = average, >1 above average
index_source	fpe_index_source	NOT NULL DEFAULT 'assumed'	assumed, historical, blended
notes	TEXT		Events or reasons for this index
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

5.5 revenue_projection_lines

Computed output of the commercial driver engine. One record per planning grain combination per revenue category.

Field	Type	Constraints	Description
line_id	UUID	PK, NOT NULL	Projection line identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference

scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
platform_id	UUID	FK -> platforms, NOT NULL	Platform scope
product_family_id	UUID	FK -> product_families, NOT NULL	Product family scope
revenue_stream_id	UUID	FK -> revenue_streams, NOT NULL	Revenue stream type
projected_orders	DECIMAL(14,4)	NOT NULL	Total projected orders for period
gross_aov_amount	DECIMAL(19,4)	NOT NULL	Gross average order value
gross_revenue_amount	DECIMAL(19,4)	NOT NULL	orders x gross_aov
platform_discount_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Platform-funded discounts
brand_discount_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Brand-funded discounts
cancellation_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Revenue lost to cancellations
net_revenue_amount	DECIMAL(19,4)	NOT NULL	Gross - discounts - cancellations
platform_commission_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Commission paid to platform
net_revenue_after_commission_amount	DECIMAL(19,4)	NOT NULL	Net revenue minus commission
vat_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	VAT collected
currency_code	CHAR(3)	NOT NULL	Currency of amounts
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6. Layer D — Cost Driver Engine

The Cost Driver Engine captures all cost assumptions. It is organized into seven tables covering the full cost structure of a dark kitchen: ingredients, packaging, labor, fixed OPEX, marketing spend, capital expenditure, and working capital policy.

Tables: unit_cost_profiles, labor_models, headcount_plans, opex_plans, marketing_plans, capex_plans, working_capital_policies

6.1 unit_cost_profiles

Per-order variable cost assumptions: food COGS, packaging, platform fees. These drive the gross margin calculation.

Field	Type	Constraints	Description
profile_id	UUID	PK, NOT NULL	Profile identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
product_family_id	UUID	FK -> product_families, NOT NULL	Product family scope
food_cost_pct	DECIMAL(5,4)	NOT NULL	Food COGS as % of net revenue
food_cost_per_order_amount	DECIMAL(19,4)		Absolute food cost per order (alternative to pct)
packaging_cost_per_order_amount	DECIMAL(19,4)	NOT NULL	Packaging cost per order
delivery_bag_cost_per_order_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Delivery bag/consumables cost
rider_tip_liability_per_order_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Tip liability if brand-paid
total_variable_cost_per_order_amount	DECIMAL(19,4)	GENERATED	Sum of all per-order variable costs
gross_margin_pct	DECIMAL(5,4)	GENERATED	1 - total_variable_cost / net_aov
food_cost_inflation_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.03	YoY food cost inflation assumption
confidence_score	DECIMAL(5,2)		
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	

is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.2 labor_models

Labor cost structure assumptions: staffing ratios, pay rates, benefit loads, and increment schedules.

Field	Type	Constraints	Description
labor_model_id	UUID	PK, NOT NULL	Labor model identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
role_name	VARCHAR(100)	NOT NULL	e.g., Kitchen Manager, Chef, Prep Cook, Packer
role_type	fpe_role_type	NOT NULL	management, production, service, admin
employment_type	fpe_employment_type	NOT NULL	full_time, part_time, contract, outsourced
headcount	DECIMAL(8,2)	NOT NULL	FTE-equivalent headcount
basic_monthly_salary_amount	DECIMAL(19,4)	NOT NULL	Monthly base salary
variable_pay_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Variable pay as % of base
benefits_load_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.15	Benefits (health, gratuity, etc.) as % of base
annual_increment_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.05	Annual salary increment
hiring_lead_time_days	SMALLINT	NOT NULL DEFAULT 30	Days from decision to start
turnover_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.25	Annual turnover rate
total_monthly_cost_amount	DECIMAL(19,4)	GENERATED	headcount x salary x (1 + benefits_load + variable)
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.3 headcount_plans

Aggregated headcount targets per period, market, and kitchen. Tracks hiring triggers and cumulative FTE.

Field	Type	Constraints	Description
plan_id	UUID	PK, NOT NULL	Plan record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
total_fte_headcount	DECIMAL(8,2)	NOT NULL	Total FTE headcount at end of period
total_monthly_payroll_amount	DECIMAL(19,4)	NOT NULL	Total payroll including all roles
hiring_trigger_orders_per_day	DECIMAL(12,4)		Order volume that triggers next hiring wave
new_hires_count	SMALLINT	NOT NULL DEFAULT 0	Net new hires in this period
departures_count	SMALLINT	NOT NULL DEFAULT 0	Departures (turnover + involuntary)
payroll_as_pct_of_revenue	DECIMAL(5,4)	GENERATED	Payroll cost / net revenue
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.4 opex_plans

Fixed and semi-fixed operating expense assumptions per kitchen per period.

Field	Type	Constraints	Description
opex_plan_id	UUID	PK, NOT NULL	OPEX plan identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period

market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NOT NULL	Kitchen scope
rent_amount	DECIMAL(19,4)	NOT NULL	Monthly rent expense
utilities_amount	DECIMAL(19,4)	NOT NULL	Electricity, water, gas
internet_telecom_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
insurance_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
cleaning_maintenance_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
technology_saas_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	POS, KDS, analytics subscriptions
platform_integration_fees_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
permits_licenses_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Annualised regulatory costs
management_fee_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	HQ allocation / management charge
other_opex_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Miscellaneous fixed costs
total_fixed_opex_amount	DECIMAL(19,4)	GENERATED	Sum of all fixed cost lines
cost_escalation_rate_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0.03	Annual escalation for fixed costs
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.5 marketing_plans

Marketing spend assumptions per period and market, with demand impact modeling.

Field	Type	Constraints	Description
marketing_plan_id	UUID	PK, NOT NULL	Plan identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope

channel_type	fpe_marketing_channel	NOT NULL	social_media, search_sem, aggregator_ads, influencer, email_sms, outdoor, events, other
budget_amount	DECIMAL(19,4)	NOT NULL	Planned marketing spend
expected_impressions	BIGINT	DEFAULT 0	Expected campaign impressions
expected_new_customers	INT	DEFAULT 0	Expected new customer acquisitions
cac_amount	DECIMAL(19,4)		Customer acquisition cost (budget / new_customers)
expected_orders_uplift_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Expected demand uplift from this channel
clv_assumption_amount	DECIMAL(19,4)		Assumed customer lifetime value
roi_target_pct	DECIMAL(5,4)		Target ROMI for this channel
spend_as_pct_of_revenue	DECIMAL(5,4)	GENERATED	budget / net_revenue
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.6 capex_plans

Capital expenditure schedule for kitchen fit-outs, equipment, technology, and expansion.

Field	Type	Constraints	Description
capex_plan_id	UUID	PK, NOT NULL	CAPEX plan identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Period of capex spend
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen (NULL = market-level)
capex_category	fpe_capex_category	NOT NULL	fit_out, equipment, technology, vehicles, working_capital, other
item_description	VARCHAR(300)	NOT NULL	Description of capex item

capex_amount	DECIMAL(19,4)	NOT NULL	Planned capex amount
useful_life_years	SMALLINT	NOT NULL DEFAULT 5	Depreciation life in years
depreciation_method	fpe_depreciation_method	NOT NULL DEFAULT 'straight_line'	straight_line, declining_balance, units_of_production
annual_depreciation_amount	DECIMAL(19,4)	GENERATED	capex / useful_life (straight-line)
is_funded	BOOLEAN	NOT NULL DEFAULT FALSE	Whether capex has committed funding
funding_source_id	UUID	FK -> equity_rounds or debt_facilities, NULLABLE	Linked funding record
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

6.7 working_capital_policies

Working capital assumptions: receivable days, payable days, inventory days, and cash buffer policies.

Field	Type	Constraints	Description
policy_id	UUID	PK, NOT NULL	Policy identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
effective_from_period_id	UUID	FK -> planning_periods, NOT NULL	Period policy becomes effective
market_id	UUID	FK -> markets, NULLABLE	NULL = company-wide policy
debtor_days	SMALLINT	NOT NULL DEFAULT 14	Platform receivable days (days to cash receipt)
creditor_days	SMALLINT	NOT NULL DEFAULT 30	Supplier payment terms
inventory_days	SMALLINT	NOT NULL DEFAULT 3	Days of food inventory held
minimum_cash_buffer_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Minimum operating cash balance
minimum_cash_buffer_months	DECIMAL(5,2)	NOT NULL DEFAULT 1.5	Alternative: months of OPEX as buffer
prepaid_deposits_amount	DECIMAL(19,4)	DEFAULT 0	Rent and utility deposits
accruals_pct_of_opex	DECIMAL(5,4)	DEFAULT 0.05	Estimated accruals buffer

notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

7. Layer E — Funding & Capital Strategy

The Funding & Capital Strategy layer models the liability and equity side of the balance sheet. It captures equity raises, debt facilities, repayment schedules, capital deployment plans, and treasury policies. Together with the cost drivers, it determines cash flow runway and capital requirements.

Tables: equity_rounds, debt_facilities, debt_repayment_schedules, capital_deployment_plans, treasury_policies

7.1 equity_rounds

Field	Type	Constraints	Description
round_id	UUID	PK, NOT NULL	Round identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
round_name	VARCHAR(100)	NOT NULL	e.g., Seed, Series A, Bridge
round_type	fpe_equity_round_type	NOT NULL	pre_seed, seed, series_a, series_b, bridge, convertible_note, strategic, secondary
target_raise_amount	DECIMAL(19,4)	NOT NULL	Target fundraise amount
expected_close_date	DATE		Expected closing date
actual_close_date	DATE		Actual closing date
pre_money_valuation_amount	DECIMAL(19,4)		Pre-money valuation
post_money_valuation_amount	DECIMAL(19,4)	GENERATED	pre_money + raise_amount
equity_dilution_pct	DECIMAL(5,4)		Ownership sold in this round
cash_received_amount	DECIMAL(19,4)		Actual cash received (net of fees)
use_of_proceeds	JSONB	DEFAULT '{}'	Allocation of proceeds by category
lead_investor	VARCHAR(200)		
status	fpe_funding_status	NOT NULL DEFAULT 'planned'	planned, in_process, closed, cancelled
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

7.2 debt_facilities

Field	Type	Constraints	Description
facility_id	UUID	PK, NOT NULL	Facility identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
facility_name	VARCHAR(100)	NOT NULL	e.g., Working Capital Line, Equipment Finance
facility_type	fpe_debt_facility_type	NOT NULL	term_loan, revolving_credit, overdraft, equipment_finance, convertible, mezzanine
lender_name	VARCHAR(200)		Lender name
facility_amount	DECIMAL(19,4)	NOT NULL	Total committed facility
drawn_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Amount currently drawn
interest_rate_pct	DECIMAL(7,6)	NOT NULL	Annual interest rate
rate_type	fpe_interest_rate_type	NOT NULL DEFAULT 'fixed'	fixed, floating, blended
start_date	DATE	NOT NULL	Facility start date
maturity_date	DATE	NOT NULL	Facility maturity
repayment_frequency	fpe_repayment_frequency	NOT NULL DEFAULT 'monthly'	monthly, quarterly, bullet
covenant_ebitda_coverage	DECIMAL(5,2)		Minimum EBITDA coverage covenant
covenant_leverage_ratio	DECIMAL(5,2)		Maximum debt/EBITDA covenant
status	fpe_facility_status	NOT NULL DEFAULT 'available'	available, drawn, repaid, cancelled, in_default
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

7.3 debt_repayment_schedules

Field	Type	Constraints	Description
schedule_id	UUID	PK, NOT NULL	Schedule record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
facility_id	UUID	FK -> debt_facilities, NOT NULL	Parent facility

period_id	UUID	FK -> planning_periods, NOT NULL	Payment period
opening_balance_amount	DECIMAL(19,4)	NOT NULL	Debt balance at period start
principal_payment_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Scheduled principal repayment
interest_payment_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Interest charge for period
total_debt_service_amount	DECIMAL(19,4)	GENERATED	principal + interest
closing_balance_amount	DECIMAL(19,4)	GENERATED	opening - principal
is_actual	BOOLEAN	NOT NULL DEFAULT FALSE	True if actual payment made
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

7.4 capital_deployment_plans

Field	Type	Constraints	Description
deployment_id	UUID	PK, NOT NULL	Deployment record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Deployment period
funding_source_id	UUID	NOT NULL	FK -> equity_rounds or debt_facilities
funding_source_type	fpe_funding_source_type	NOT NULL	equity, debt
deployment_category	fpe_capex_category	NOT NULL	Aligns to capex category
planned_amount	DECIMAL(19,4)	NOT NULL	Planned deployment amount
actual_amount	DECIMAL(19,4)	DEFAULT 0	Actual deployed amount
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

7.5 treasury_policies

Field	Type	Constraints	Description
policy_id	UUID	PK, NOT NULL	Policy identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
effective_from_period_id	UUID	FK -> planning_periods, NOT NULL	
minimum_cash_runway_months	DECIMAL(5,2)	NOT NULL DEFAULT 6	Trigger for fundraising / cost action
target_cash_runway_months	DECIMAL(5,2)	NOT NULL DEFAULT 12	Target operating buffer
fx_hedging_policy	TEXT		FX risk mitigation approach
interest_rate_policy	TEXT		Fixed vs floating preference
cash_sweep_threshold_amount	DECIMAL(19,4)	DEFAULT 0	Above this balance, sweep to short-term investments
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

8. Layer F — Financial Projection Engine

The Financial Projection Engine aggregates all driver outputs into the three core financial statements: Profit & Loss, Cash Flow Statement, and Balance Sheet. It also computes unit economics — the per-order, per-kitchen, and per-market financial profile.

Tables: pnl_projections, cashflow_projections, balance_sheet_projections, unit_economics_projections

P&L Structure: Net Revenue → Gross Profit → Contribution Margin → EBITDA → EBIT → EBT → Net Profit

8.1 pnl_projections

Full projected income statement per planning period. Aggregated from commercial and cost driver outputs.

Field	Type	Constraints	Description
pnl_id	UUID	PK, NOT NULL	P&L record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NULLABLE	NULL = consolidated
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen-level P&L
gross_revenue_amount	DECIMAL(19,4)	NOT NULL	Gross order value before all deductions
platform_discounts_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Platform-funded discounts
brand_discounts_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Brand-funded discounts
cancellations_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Revenue lost to cancellations
net_revenue_amount	DECIMAL(19,4)	NOT NULL	Gross - discounts - cancellations
platform_commission_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Commission paid to platforms
net_revenue_after_commission_amount	DECIMAL(19,4)	NOT NULL	Net revenue minus commission
food_cogs_amount	DECIMAL(19,4)	NOT NULL	Food ingredient costs
packaging_cogs_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Packaging costs
other_cogs_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Other variable costs
total_cogs_amount	DECIMAL(19,4)	NOT NULL	Total cost of goods sold
gross_profit_amount	DECIMAL(19,4)	GENERATED	Net revenue after commission - COGS

gross_margin_pct	DECIMAL(5,4)	GENERATED	gross_profit / net_revenue
payroll_amount	DECIMAL(19,4)	NOT NULL	Total labor cost
rent_amount	DECIMAL(19,4)	NOT NULL	Rent expense
utilities_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
marketing_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
technology_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
other_opex_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
total_opex_amount	DECIMAL(19,4)	GENERATED	Sum of all opex lines
ebitda_amount	DECIMAL(19,4)	GENERATED	Gross profit - total opex
ebitda_margin_pct	DECIMAL(5,4)	GENERATED	ebitda / net_revenue
depreciation_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	D&A from capex schedule
amortization_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
ebit_amount	DECIMAL(19,4)	GENERATED	EBITDA - D&A
interest_expense_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Interest on debt facilities
interest_income_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Interest on cash balances
ebt_amount	DECIMAL(19,4)	GENERATED	EBIT - interest net
tax_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Corporate tax provision
net_profit_amount	DECIMAL(19,4)	GENERATED	EBT - tax
net_margin_pct	DECIMAL(5,4)	GENERATED	net_profit / net_revenue
total_orders	DECIMAL(14,4)	NOT NULL	Total orders in period
average_order_value_amount	DECIMAL(19,4)	GENERATED	net_revenue / total_orders
currency_code	CHAR(3)	NOT NULL	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

8.2 cashflow_projections

Projected cash flow statement: operating, investing, and financing activities. Derives cash runway.

Field	Type	Constraints	Description
cf_id	UUID	PK, NOT NULL	Cash flow record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context

period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NULLABLE	Market scope or consolidated
opening_cash_amount	DECIMAL(19,4)	NOT NULL	Cash balance at period start
ebitda_amount	DECIMAL(19,4)	NOT NULL	From P&L projection
working_capital_movement_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Change in WC (debtor/creditor/inventory)
tax_paid_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Cash tax payments
operating_cash_flow_amount	DECIMAL(19,4)	GENERATED	EBITDA +/- WC - tax
capex_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Capital expenditure payments
asset_disposal_proceeds_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
investing_cash_flow_amount	DECIMAL(19,4)	GENERATED	-capex + disposals
equity_raised_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Equity inflows
debt_drawn_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Debt drawdowns
debt_repaid_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Principal repayments
interest_paid_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
dividends_paid_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
financing_cash_flow_amount	DECIMAL(19,4)	GENERATED	equity + debt drawn - repaid - interest - dividends
net_cash_movement_amount	DECIMAL(19,4)	GENERATED	operating + investing + financing
closing_cash_amount	DECIMAL(19,4)	GENERATED	opening + net movement
cash_runway_months	DECIMAL(8,2)	GENERATED	closing_cash / monthly_burn_rate
monthly_burn_rate_amount	DECIMAL(19,4)	GENERATED	Net cash used in operations if negative
currency_code	CHAR(3)	NOT NULL	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

8.3 balance_sheet_projections

Projected balance sheet: assets, liabilities, and equity. Validates cash flow projections via balance sheet identity.

Field	Type	Constraints	Description
bs_id	UUID	PK, NOT NULL	Balance sheet record identifier

company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Balance sheet date (period end)
cash_and_equivalents_amount	DECIMAL(19,4)	NOT NULL	
accounts_receivable_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Platform payables owed to company
inventory_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Food and packaging inventory
prepaid_expenses_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Prepaid rent, deposits
other_current_assets_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
total_current_assets_amount	DECIMAL(19,4)	GENERATED	Sum of current assets
gross_ppe_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Gross property, plant & equipment
accumulated_depreciation_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
net_ppe_amount	DECIMAL(19,4)	GENERATED	gross_ppe - accum_depreciation
intangible_assets_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Technology, licenses
other_non_current_assets_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
total_assets_amount	DECIMAL(19,4)	GENERATED	total_current + net_ppe + intangibles + other_nc
accounts_payable_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Supplier payables
accrued_liabilities_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Accruals
deferred_revenue_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Subscriptions / advance payments
current_portion_of_debt_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
other_current_liabilities_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
total_current_liabilities_amount	DECIMAL(19,4)	GENERATED	
long_term_debt_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
other_non_current_liabilities_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	
total_liabilities_amount	DECIMAL(19,4)	GENERATED	
paid_in_capital_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Equity raised to date

retained_earnings_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Cumulative net profit
total_equity_amount	DECIMAL(19,4)	GENERATED	paid_in + retained_earnings
total_liabilities_and_equity_amount	DECIMAL(19,4)	GENERATED	Must equal total_assets
balance_check	BOOLEAN	GENERATED	total_assets = total_liabilities_and_equity
currency_code	CHAR(3)	NOT NULL	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

8.4 unit_economics_projections

Per-order and per-kitchen unit economics. Key decision metrics for expansion, pricing, and investment.

Field	Type	Constraints	Description
ue_id	UUID	PK, NOT NULL	Unit economics record
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
total_orders	DECIMAL(14,4)	NOT NULL	Total orders in period
orders_per_day	DECIMAL(12,4)	GENERATED	total_orders / trading_days
net_revenue_per_order_amount	DECIMAL(19,4)	GENERATED	net_revenue / total_orders
cogs_per_order_amount	DECIMAL(19,4)	GENERATED	total_cogs / total_orders
gross_profit_per_order_amount	DECIMAL(19,4)	GENERATED	gross_profit / total_orders
contribution_margin_per_order_amount	DECIMAL(19,4)	GENERATED	gross_profit - variable opex per order
ebitda_per_order_amount	DECIMAL(19,4)	GENERATED	ebitda / total_orders
monthly_revenue_per_kitchen_amount	DECIMAL(19,4)	GENERATED	net_revenue / active_kitchens
monthly_ebitda_per_kitchen_amount	DECIMAL(19,4)	GENERATED	ebitda / active_kitchens

revenue_per_sqm_amount	DECIMAL(19,4)	GENERATED	net_revenue / total_sqm
ebitda_per_sqm_amount	DECIMAL(19,4)	GENERATED	ebitda / total_sqm
payback_period_months	DECIMAL(8,2)	GENERATED	total_capex / monthly_ebitda
roic_pct	DECIMAL(7,4)	GENERATED	annualised_ebitda / invested_capital
capacity_utilization_pct	DECIMAL(5,4)	GENERATED	actual_orders / max_capacity_orders
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

9. Layer G — KPI & Performance Intelligence

The KPI and Performance Intelligence layer captures the key performance indicators computed from the financial projections, enables scenario comparison, provides driver explainability, and triggers alerts when KPIs breach thresholds.

Tables: kpi_projections, scenario_comparison_snapshots, driver_explainability_logs, performance_alerts

9.1 kpi_projections

Field	Type	Constraints	Description
kpi_id	UUID	PK, NOT NULL	KPI record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NULLABLE	Market scope or consolidated
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
kpi_name	VARCHAR(100)	NOT NULL	e.g., ebitda_margin_pct, cash_runway_months, irr_pct
kpi_category	fpe_kpi_category	NOT NULL	revenue, profitability, liquidity, growth, efficiency, return
kpi_value	DECIMAL(19,4)	NOT NULL	Computed KPI value
kpi_unit	VARCHAR(30)	NOT NULL	e.g., %, AED, months, x
target_value	DECIMAL(19,4)		Target KPI value for this period
prior_period_value	DECIMAL(19,4)		Prior period KPI for comparison
variance_from_target	DECIMAL(19,4)	GENERATED	kpi_value - target_value
variance_pct_from_target	DECIMAL(7,4)	GENERATED	variance / target_value
rag_status	fpe_rag_status	GENERATED	red, amber, green based on thresholds
irr_pct	DECIMAL(7,4)		Internal rate of return (annualised)
npv_amount	DECIMAL(19,4)		Net present value
wacc_pct	DECIMAL(7,4)		Weighted average cost of capital used
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	

is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

9.2 scenario_comparison_snapshots

Field	Type	Constraints	Description
snapshot_id	UUID	PK, NOT NULL	Snapshot identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
base_scenario_id	UUID	FK -> scenarios, NOT NULL	Reference scenario
comparison_scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario being compared
period_id	UUID	FK -> planning_periods, NULLABLE	NULL = full horizon comparison
comparison_type	fpe_comparison_type	NOT NULL	budget_vs_forecast, base_vs_upside, base_vs_stress, forecast_vs_actuals
kpi_name	VARCHAR(100)	NOT NULL	KPI being compared
base_value	DECIMAL(19,4)	NOT NULL	Base scenario value
comparison_value	DECIMAL(19,4)	NOT NULL	Comparison scenario value
absolute_variance	DECIMAL(19,4)	GENERATED	comparison - base
relative_variance_pct	DECIMAL(7,4)	GENERATED	absolute / base
is_material	BOOLEAN	GENERATED	True if abs variance > materiality threshold
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

9.3 driver_explainability_logs

Traces the contribution of each driver to changes in KPIs between periods or scenarios. Powers bridge/waterfall charts.

Field	Type	Constraints	Description
log_id	UUID	PK, NOT NULL	Log entry identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
period_id	UUID	FK -> planning_periods, NOT NULL	Period of analysis
kpi_name	VARCHAR(100)	NOT NULL	Target KPI being explained
driver_table	VARCHAR(100)	NOT NULL	Source table of driver (e.g., demand_drivers)

driver_field	VARCHAR(100)	NOT NULL	Specific field driving the change
driver_label	VARCHAR(200)	NOT NULL	Human-readable label for the driver
contribution_amount	DECIMAL(19,4)	NOT NULL	Absolute contribution to KPI change
contribution_pct	DECIMAL(7,4)	NOT NULL	% share of total change
from_value	DECIMAL(19,4)		Driver value in prior period/scenario
to_value	DECIMAL(19,4)		Driver value in current period/scenario
display_order	SMALLINT	NOT NULL DEFAULT 0	Order in waterfall chart
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

9.4 performance_alerts

Field	Type	Constraints	Description
alert_id	UUID	PK, NOT NULL	Alert identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
period_id	UUID	FK -> planning_periods, NOT NULL	Period of alert
market_id	UUID	FK -> markets, NULLABLE	Market scope
alert_type	fpe_alert_type	NOT NULL	kpi_breach, runway_warning, covenant_breach, capex_overrun, assumption_stale
alert_severity	fpe_alert_severity	NOT NULL	info, warning, critical
kpi_name	VARCHAR(100)		KPI triggering alert
threshold_value	DECIMAL(19,4)		Breach threshold
actual_value	DECIMAL(19,4)		Actual/projected value that triggered alert
alert_message	TEXT	NOT NULL	Human-readable alert description
triggered_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
acknowledged_by	UUID		User who acknowledged alert
acknowledged_at	TIMESTAMPTZ		
resolution_notes	TEXT		
is_resolved	BOOLEAN	NOT NULL DEFAULT FALSE	
resolved_at	TIMESTAMPTZ		

created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

10. Layer H — Risk & Uncertainty Framework

The Risk & Uncertainty Framework provides a structured approach to identifying, quantifying, and managing financial risks. It enables uncertainty bands on all projections, stress testing, scenario-based risk quantification, and trigger-based risk monitoring.

Tables: risk_scenarios, risk_registers, uncertainty_bands, stress_test_definitions, risk_trigger_log

10.1 risk_scenarios

Field	Type	Constraints	Description
risk_scenario_id	UUID	PK, NOT NULL	Risk scenario identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
parent_scenario_id	UUID	FK -> scenarios, NOT NULL	Planning scenario this risk applies to
risk_scenario_name	VARCHAR(200)	NOT NULL	e.g., Platform Commission Increase +5%, Ramadan Demand Spike
risk_category	fpe_risk_category	NOT NULL	demand, pricing, cost, platform, regulatory, macro, operational, liquidity
probability_pct	DECIMAL(5,4)	NOT NULL CHECK (0<=x<=1)	Probability of scenario occurring
impact_on_ebitda_amount	DECIMAL(19,4)	NOT NULL	Expected EBITDA impact if scenario occurs
impact_on_cash_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Expected cash impact
expected_value_impact_amount	DECIMAL(19,4)	GENERATED	probability x impact
time_horizon_months	SMALLINT	NOT NULL DEFAULT 12	How far out this risk applies
mitigation_actions	TEXT		Planned mitigants
residual_probability_pct	DECIMAL(5,4)		Post-mitigation probability
residual_impact_amount	DECIMAL(19,4)		Post-mitigation EBITDA impact
status	fpe_risk_status	NOT NULL DEFAULT 'identified'	identified, mitigated, materialised, closed
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

10.2 risk_registers

Field	Type	Constraints	Description
register_id	UUID	PK, NOT NULL	Register entry identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
risk_name	VARCHAR(200)	NOT NULL	Descriptive risk name
risk_description	TEXT	NOT NULL	Full risk description
risk_category	fpe_risk_category	NOT NULL	
risk_owner_id	UUID	NOT NULL	User responsible for managing this risk
likelihood_score	SMALLINT	CHECK (1<=x<=5)	1=Rare to 5=Almost Certain
impact_score	SMALLINT	CHECK (1<=x<=5)	1=Negligible to 5=Catastrophic
risk_score	SMALLINT	GENERATED	likelihood x impact
risk_rating	fpe_risk_rating	GENERATED	low(1-5), medium(6-12), high(13-19), critical(20-25)
control_effectiveness	fpe_control_effectiveness	DEFAULT 'partial'	full, partial, none
review_frequency_days	SMALLINT	NOT NULL DEFAULT 30	
last_review_date	DATE		
next_review_date	DATE		
status	fpe_risk_status	NOT NULL DEFAULT 'open'	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

10.3 uncertainty_bands

Defines confidence intervals and uncertainty ranges for KPI projections. Supports probabilistic financial planning.

Field	Type	Constraints	Description
band_id	UUID	PK, NOT NULL	Band record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
market_id	UUID	FK -> markets, NULLABLE	Market scope

kpi_name	VARCHAR(100)	NOT NULL	KPI being bounded
p10_value	DECIMAL(19,4)	NOT NULL	10th percentile (pessimistic)
p25_value	DECIMAL(19,4)	NOT NULL	25th percentile
p50_value	DECIMAL(19,4)	NOT NULL	50th percentile (median/base)
p75_value	DECIMAL(19,4)	NOT NULL	75th percentile
p90_value	DECIMAL(19,4)	NOT NULL	90th percentile (optimistic)
methodology	fpe_band_methodology	NOT NULL DEFAULT 'assumed'	assumed, historical, monte_carlo, expert_judgment
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

10.4 stress_test_definitions

Field	Type	Constraints	Description
stress_test_id	UUID	PK, NOT NULL	Stress test identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
test_name	VARCHAR(200)	NOT NULL	e.g., Red Pill: -30% Demand + 10% Cost Shock
test_description	TEXT	NOT NULL	Full narrative description
demand_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in demand vs base
aov_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in AOV vs base
food_cost_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in food costs vs base
platform_commission_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in platform commission
rent_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in rent
marketing_spend_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	% change in marketing budget
fx_shock_pct	DECIMAL(7,4)	NOT NULL DEFAULT 0	FX rate change if multi-currency
is_regulatory_scenario	BOOLEAN	NOT NULL DEFAULT FALSE	True for regulatory stress tests
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

10.5 risk_trigger_log

Field	Type	Constraints	Description
trigger_id	UUID	PK, NOT NULL	Trigger log identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
risk_register_id	UUID	FK -> risk_registers, NOT NULL	Associated risk
trigger_type	fpe_trigger_type	NOT NULL	kpi_threshold, time_based, event_based, manual
triggered_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
trigger_condition	TEXT	NOT NULL	Condition that triggered (e.g., EBITDA < 0)
trigger_value	DECIMAL(19,4)		Value at trigger time
threshold_value	DECIMAL(19,4)		Threshold that was breached
recommended_action	TEXT		System-recommended response
action_taken	TEXT		Actual action taken
actioned_by	UUID		
actioned_at	TIMESTAMPTZ		
is_resolved	BOOLEAN	NOT NULL DEFAULT FALSE	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

11. Layer I — Simulation & Decision Support

The Simulation & Decision Support layer enables analysts to run structured what-if analyses, sensitivity calculations, Monte Carlo simulations, breakeven analysis, and maintain a decision memory for institutional learning.

Tables: simulation_runs, sensitivity_analyses, monte_carlo_outputs, breakeven_analyses, decision_memory

11.1 simulation_runs

Field	Type	Constraints	Description
run_id	UUID	PK, NOT NULL	Simulation run identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
base_scenario_id	UUID	FK -> scenarios, NOT NULL	Base scenario for simulation
run_name	VARCHAR(200)	NOT NULL	e.g., Expansion Pace Simulator Q3 2025
simulation_type	fpe_simulation_type	NOT NULL	pricing, menu_mix, expansion_pace, marketing_spend, capex_timing, cost_shock, platform_dependency, staffing, scenario_compare, sensitivity, monte_carlo, breakeven, capital_requirements
run_by	UUID	NOT NULL	User who triggered the run
run_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	Run timestamp
parameter_overrides	JSONB	NOT NULL DEFAULT '{}'	Driver overrides applied in this run
run_status	fpe_run_status	NOT NULL DEFAULT 'queued'	queued, running, completed, failed
started_at	TIMESTAMPTZ		
completed_at	TIMESTAMPTZ		
error_message	TEXT		Error details if failed
result_summary	JSONB		Key output metrics in JSON
notes	TEXT		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

11.2 sensitivity_analyses

One-way and two-way sensitivity outputs. Each record captures the impact of one driver change on one KPI.

Field	Type	Constraints	Description
analysis_id	UUID	PK, NOT NULL	Analysis record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
run_id	UUID	FK -> simulation_runs, NOT NULL	Parent simulation run
driver_name	VARCHAR(200)	NOT NULL	Driver being varied
driver_table	VARCHAR(100)	NOT NULL	Source table of driver
driver_field	VARCHAR(100)	NOT NULL	Field being varied
driver_change_pct	DECIMAL(7,4)	NOT NULL	% change from base applied
target_kpi	VARCHAR(100)	NOT NULL	KPI being measured
base_kpi_value	DECIMAL(19,4)	NOT NULL	Base scenario KPI value
stressed_kpi_value	DECIMAL(19,4)	NOT NULL	Stressed KPI value
kpi_change_amount	DECIMAL(19,4)	GENERATED	stressed - base
kpi_elasticity	DECIMAL(8,6)	GENERATED	% KPI change / % driver change
sensitivity_rank	SMALLINT		Rank 1=most sensitive (for tornado chart)
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

11.3 monte_carlo_outputs

Field	Type	Constraints	Description
output_id	UUID	PK, NOT NULL	Output record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
run_id	UUID	FK -> simulation_runs, NOT NULL	Parent simulation run
iteration_number	INT	NOT NULL	Simulation iteration (1 to N)
kpi_name	VARCHAR(100)	NOT NULL	KPI output
kpi_value	DECIMAL(19,4)	NOT NULL	KPI value in this iteration
driver_values	JSONB		Driver values used in this iteration
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

11.4 breakeven_analyses

Field	Type	Constraints	Description
be_id	UUID	PK, NOT NULL	Breakeven record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NOT NULL	Scenario context
assumption_set_id	UUID	FK -> assumption_sets, NOT NULL	Assumption set context
period_id	UUID	FK -> planning_periods, NOT NULL	Period of analysis
market_id	UUID	FK -> markets, NOT NULL	Market scope
kitchen_id	UUID	FK -> kitchens, NULLABLE	Kitchen scope
breakeven_metric	fpe_breakeven_metric	NOT NULL	ebitda, gross_profit, cash_flow, unit_economics
fixed_costs_amount	DECIMAL(19,4)	NOT NULL	Total fixed cost base
contribution_margin_per_order_amount	DECIMAL(19,4)	NOT NULL	Contribution margin per order
breakeven_orders_per_day	DECIMAL(12,4)	GENERATED	fixed_costs / (trading_days x cm_per_order)
breakeven_revenue_amount	DECIMAL(19,4)	GENERATED	breakeven_orders x net_aov
actual_orders_per_day	DECIMAL(12,4)	NOT NULL	Actual/projected daily orders
margin_of_safety_pct	DECIMAL(5,4)	GENERATED	(actual - breakeven) / actual
months_to_breakeven	DECIMAL(8,2)		Estimated months to reach breakeven at current ramp
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

11.5 decision_memory

Captures key financial decisions made, the evidence base used, and actual outcomes. Enables institutional learning and decision audit trails.

Field	Type	Constraints	Description
decision_id	UUID	PK, NOT NULL	Decision record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_id	UUID	FK -> scenarios, NULLABLE	Scenario in which decision was made

decision_type	fpe_decision_type	NOT NULL	expansion_approval, capex_commitment, pricing_change, hiring_trigger, fundraise_initiation, scenario_freeze, opex_cut
decision_title	VARCHAR(300)	NOT NULL	Short title of the decision
decision_description	TEXT	NOT NULL	Full narrative
market_id	UUID	FK -> markets, NULLABLE	Relevant market
kitchen_id	UUID	FK -> kitchens, NULLABLE	Relevant kitchen
financial_impact_amount	DECIMAL(19,4)		Expected financial impact
confidence_at_decision	DECIMAL(5,2)		Confidence score at time of decision
supporting_kpis	JSONB	DEFAULT '{}'	KPI snapshot at decision time
decision_made_by	UUID	NOT NULL	Primary decision maker
decision_date	DATE	NOT NULL	Date of decision
review_date	DATE		Date to review outcome
actual_outcome_notes	TEXT		Post-decision outcome assessment
outcome_assessment	fpe_outcome_assessment		better_than_expected, as_expected, worse_than_expected
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

12. Layer J — Portfolio Construction

The Portfolio Construction layer enables multi-market, multi-kitchen financial planning at the portfolio level. It captures portfolio scenarios, market allocations, cross-market synergies, and capital allocation decisions across the full geographic footprint.

Tables: portfolio_scenarios, market_portfolio_allocations, cross_market_synergies, portfolio_capital_allocations

12.1 portfolio_scenarios

Field	Type	Constraints	Description
portfolio_scenario_id	UUID	PK, NOT NULL	Portfolio scenario identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
scenario_name	VARCHAR(200)	NOT NULL	e.g., Conservative Expansion 2025-2027
portfolio_strategy	fpe_portfolio_strategy	NOT NULL	concentrate, diversify, accelerate, optimize, harvest
target_market_count	SMALLINT	NOT NULL DEFAULT 1	Target number of active markets
target_kitchen_count	SMALLINT	NOT NULL DEFAULT 1	Target number of active kitchens
planning_horizon_months	SMALLINT	NOT NULL DEFAULT 36	Portfolio horizon (typically longer than unit plan)
total_capex_budget_amount	DECIMAL(19,4)	NOT NULL	Total capex envelope for portfolio
target_portfolio_irr_pct	DECIMAL(7,4)		Target blended IRR across all markets
target_portfolio_ebitda_margin_pct	DECIMAL(5,4)		Target consolidated EBITDA margin
status	fpe_scenario_status	NOT NULL DEFAULT 'draft'	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
created_by	UUID	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

12.2 market_portfolio_allocations

Field	Type	Constraints	Description
allocation_id	UUID	PK, NOT NULL	Allocation record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference

portfolio_scenario_id	UUID	FK -> portfolio_scenarios, NOT NULL	Parent portfolio scenario
market_id	UUID	FK -> markets, NOT NULL	Market being allocated to
period_id	UUID	FK -> planning_periods, NOT NULL	Planning period
allocated_capex_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Capex allocated to this market
allocated_opex_budget_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	OPEX budget allocated
allocated_marketing_budget_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Marketing budget allocated
revenue_contribution_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Expected % of total portfolio revenue
ebitda_contribution_pct	DECIMAL(5,4)	NOT NULL DEFAULT 0	Expected % of total portfolio EBITDA
strategic_priority	fpe_strategic_priority	NOT NULL DEFAULT 'medium'	high, medium, low, exit
expansion_phase	fpe_expansion_phase	NOT NULL DEFAULT 'planned'	planned, active, scaling, optimising, exiting
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

12.3 cross_market_synergies

Field	Type	Constraints	Description
synergy_id	UUID	PK, NOT NULL	Synergy record identifier
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
portfolio_scenario_id	UUID	FK -> portfolio_scenarios, NOT NULL	Parent portfolio scenario
market_a_id	UUID	FK -> markets, NOT NULL	First market in synergy pair
market_b_id	UUID	FK -> markets, NOT NULL	Second market in synergy pair
synergy_type	fpe_synergy_type	NOT NULL	supply_chain, brand_spill, shared_tech, shared_ops, shared_marketing, economies_of_scale
synergy_description	TEXT	NOT NULL	Description of synergy mechanism
annual_cost_saving_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Annual cost saving from synergy
annual_revenue_uplift_amount	DECIMAL(19,4)	NOT NULL DEFAULT 0	Revenue uplift from synergy

realisation_timeline_months	SMALLINT	NOT NULL DEFAULT 6	Months to fully realise synergy
confidence_score	DECIMAL(5,2)		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

12.4 portfolio_capital_allocations

Field	Type	Constraints	Description
cap_alloc_id	UUID	PK, NOT NULL	Capital allocation record
company_id	UUID	FK -> companies, NOT NULL	Tenant reference
portfolio_scenario_id	UUID	FK -> portfolio_scenarios, NOT NULL	Parent portfolio scenario
market_id	UUID	FK -> markets, NOT NULL	Market receiving capital
period_id	UUID	FK -> planning_periods, NOT NULL	Allocation period
funding_source_id	UUID	NOT NULL	FK -> equity_rounds or debt_facilities
funding_source_type	fpe_funding_source_type	NOT NULL	equity, debt
allocated_amount	DECIMAL(19,4)	NOT NULL	Capital allocated to this market in this period
allocation_purpose	fpe_capex_category	NOT NULL	fit_out, equipment, working_capital, marketing, other
expected_irr_pct	DECIMAL(7,4)		Expected IRR from this market allocation
expected_payback_months	DECIMAL(8,2)		Expected payback period
approval_status	fpe_approval_status	NOT NULL DEFAULT 'pending'	
approved_by	UUID		
approved_at	TIMESTAMPTZ		
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
is_deleted	BOOLEAN	NOT NULL DEFAULT FALSE	

13. Optional Extension Layer — Operational Intelligence

The Optional Extension Layer contains 11 tables that capture granular operational detail beyond the 6-to-18-month finance-first horizon. These tables are deferred to Phase 6+ of the engineering roadmap. They are designed to integrate seamlessly with the core schema but are not required for the finance-first planning engine to function.

13.1 Scope & Rationale

The extension layer is triggered when the business reaches the Steady State planning grain (Kitchen x Platform x Product Family x Month) and requires: (1) actuals ingestion from POS/platform APIs; (2) intra-day demand forecasting for kitchen scheduling; (3) detailed SKU-level recipe and costing; (4) kitchen station capacity modelling; (5) workforce scheduling optimization.

Table	Purpose	Deferred Reason
order_headers	Raw delivery order records from platform APIs	Not needed for driver-based projections
order_lines	Line items within each order (SKU level)	SKU detail not in planning grain
product_variants	Individual SKUs, sizes, modifiers with pricing	Product family sufficient for planning
product_modifiers	Add-on options and modifier groups	Below planning grain boundary
ingredient_bom	Bill of materials per product variant	Recipe detail is operational extension
ingredients	Ingredient master: supplier, unit cost, shelf life	Supply chain detail beyond planning scope
kitchen_stations	Station definitions: oven, prep, packing, assembly	Station-level capacity is operational
intraday_demand_forecasts	30-minute bucket demand forecasts per kitchen	Sub-daily grain is operational scheduling
intraday_labor_schedules	Shift-level staffing plans with 30-min slots	Scheduling detail beyond planning horizon
actuals_revenue_bridge	Actuals vs forecast bridge at order level	Requires actuals ingestion pipeline
actuals_cost_bridge	Actuals vs forecast cost variance at invoice level	Requires ERP/accounting integration

13.2 Extension Integration Principles

- All extension tables inherit the same company_id, scenario_id, planning_period_id reference structure.
- Extension tables reference core Layer B entities (markets, kitchens, platforms, product_families) to maintain schema integrity.
- Actuals ingested via extension tables feed into the planning_audit_log and can update is_actuals_available on planning_periods.
- Extension tables use the same UUID PK convention, audit fields, and soft-delete pattern as core tables.
- When actuals are available, the Actuals Bridge tables enable budget-vs-actuals variance analysis without modifying projection records.

14. Indexes, Performance & Partitioning

14.1 Core Composite Indexes

All tables carrying the planning grain key should have a composite index. Key indexes:

Index Name	Table	Columns	Type	Purpose
idx_pnl_grain	pnl_projections	company_id, scenario_id, assumption_set_id, period_id, market_id	BTREE	Primary grain lookup for P&L
idx_cf_grain	cashflow_projections	company_id, scenario_id, assumption_set_id, period_id	BTREE	Primary grain lookup for CF
idx_demand_grain	demand_drivers	company_id, scenario_id, assumption_set_id, period_id, market_id, platform_id, product_family_id	BTREE	Full demand grain
idx_price_grain	price_plans	company_id, scenario_id, assumption_set_id, period_id, market_id, platform_id, product_family_id	BTREE	Full price grain
idx_ue_grain	unit_economics_projections	company_id, scenario_id, period_id, market_id, kitchen_id	BTREE	Unit economics grain
idx_kpi_name	kpi_projections	company_id, scenario_id, period_id, kpi_name	BTREE	KPI lookup by name
idx_scenarios_company	scenarios	company_id, status, is_deleted	BTREE	Active scenario listing
idx_periods_calendar	planning_periods	calendar_id, period_start_date, period_end_date	BTREE	Period range queries
idx_markets_hierarchy	markets	company_id, parent_market_id, market_level	BTREE	Geographic hierarchy traversal
idx_risk_trigger_log	risk_trigger_log	company_id, scenario_id, triggered_at	BTREE	Risk trigger timeline
idx_audit_log_record	planning_audit_log	company_id, table_name, record_id, changed_at	BTREE	Audit trail by record
idx_simulation_runs_type	simulation_runs	company_id, base_scenario_id, simulation_type, run_status	BTREE	Simulation filtering

14.2 Partial Indexes

All tenant queries should use partial indexes on non-deleted records:

Pattern	Example
Active records only	CREATE INDEX idx_scenarios_active ON scenarios(company_id, status) WHERE is_deleted = FALSE;
Published plan versions	CREATE INDEX idx_versions_published ON plan_versions(company_id, scenario_id) WHERE is_published = TRUE;
Active kitchens	CREATE INDEX idx_kitchens_active ON kitchens(company_id, market_id) WHERE is_deleted = FALSE AND status = 'operational';
Unresolved alerts	CREATE INDEX idx_alerts_open ON performance_alerts(company_id, alert_severity, triggered_at) WHERE is_resolved = FALSE;

14.3 Partitioning Strategy

Table	Partition Key	Strategy	Rationale
pnl_projections	company_id	HASH (8 partitions)	Tenant isolation for large multi-company deployments
cashflow_projections	company_id	HASH (8 partitions)	Same as P&L
planning_audit_log	changed_at	RANGE (monthly)	Audit logs grow indefinitely; range partition enables archival
monte_carlo_outputs	run_id	HASH (16 partitions)	MC generates millions of rows; distribute evenly
driver_explainability_logs	company_id, period_id	HASH + RANGE	Time-series plus tenant distribution

15. Enum Reference

All PostgreSQL enum types used in the schema. Enum names are prefixed with fpe_.

Enum Type	Values
fpe_scenario_type	base, upside, downside, stress, budget, forecast, actuals, portfolio
fpe_scenario_status	draft, in_review, approved, published, archived, frozen
fpe_approval_status	pending, submitted, approved, rejected, escalated
fpe_period_type	daily, weekly, monthly, quarterly, semi_annual, annual
fpe_market_level	country, cluster, macro, micro
fpe_market_status	prospective, approved, active, paused, exited
fpe_kitchen_status	planned, under_fit_out, soft_launch, operational, paused, closed
fpe_platform_type	third_party_aggregator, owned_app, owned_web, direct_phone, b2b, catering
fpe_product_family_type	core, complementary, premium, limited_time, b2b, seasonal
fpe_revenue_stream_type	platform_delivery, own_channel, catering, b2b, subscription, other
fpe_revenue_recognition	on_delivery, on_payment, ratable, milestone
fpe_capex_category	fit_out, equipment, technology, vehicles, working_capital, permits, other
fpe_depreciation_method	straight_line, declining_balance, units_of_production, sum_of_years
fpe_equity_round_type	pre_seed, seed, series_a, series_b, series_c, bridge, convertible_note, strategic, secondary
fpe_debt_facility_type	term_loan, revolving_credit, overdraft, equipment_finance, convertible, mezzanine
fpe_interest_rate_type	fixed, floating, blended
fpe_repayment_frequency	monthly, quarterly, semi_annual, annual, bullet, custom
fpe_funding_status	planned, in_process, closed, cancelled, extended
fpe_kpi_category	revenue, profitability, liquidity, growth, efficiency, return, operational
fpe_rag_status	green, amber, red, grey
fpe_alert_type	kpi_breach, runway_warning, covenant_breach, capex_overrun, assumption_stale, expansion_trigger, risk_materialised
fpe_alert_severity	info, warning, critical, emergency
fpe_risk_category	demand, pricing, cost, platform, regulatory, macro, operational, liquidity, competitive, reputational
fpe_risk_status	identified, assessed, mitigated, materialised, closed
fpe_risk_rating	low, medium, high, critical

fpe_simulation_type	pricing, menu_mix, expansion_pace, marketing_spend, capex_timing, cost_shock, platform_dependency, staffing, scenario_compare, sensitivity, monte_carlo, breakeven, capital_requirements
fpe_run_status	queued, running, completed, failed, cancelled
fpe_breakeven_metric	ebitda, gross_profit, cash_flow, contribution_margin, net_profit
fpe_decision_type	expansion_approval, capex_commitment, pricing_change, hiring_trigger, fundraise_initiation, scenario_freeze, opex_cut, market_exit
fpe_outcome_assessment	better_than_expected, as_expected, worse_than_expected, not_yet_assessed
fpe_portfolio_strategy	concentrate, diversify, accelerate, optimize, harvest
fpe_strategic_priority	high, medium, low, exit
fpe_expansion_phase	planned, approved, active, scaling, optimising, exiting
fpe_synergy_type	supply_chain, brand_spill, shared_tech, shared_ops, shared_marketing, economies_of_scale
fpe_volatility_level	low, medium, high, very_high
fpe_governance_entity_type	scenario, assumption_set, plan_version, driver_record, projection_record
fpe_governance_event_type	submitted, approved, rejected, frozen, published, escalated, recalled, amended
fpe_audit_operation	INSERT, UPDATE, DELETE, TRUNCATE
fpe_marketing_channel	social_media, search_sem, aggregator_ads, influencer, email_sms, outdoor, events, loyalty, referral, other
fpe_role_type	management, production, service, admin, tech, finance
fpe_employment_type	full_time, part_time, contract, outsourced, intern
fpe_menu_node_type	menu, collection, category, product_family
fpe_expansion_recommendation	enter, monitor, defer, avoid, exit
fpe_calendar_type	gregorian, fiscal, tax, custom
fpe_band_methodology	assumed, historical, monte_carlo, expert_judgment, blended
fpe_index_source	assumed, historical, blended, third_party
fpe_funding_source_type	equity, debt, grant, internal
fpe_comparison_type	budget_vs_forecast, base_vs_upside, base_vs_downside, base_vs_stress, forecast_vs_actuals, period_over_period
fpe_trigger_type	kpi_threshold, time_based, event_based, manual, system_automated
fpe_control_effectiveness	full, partial, none
fpe_assumption_set_type	budget, forecast, actuals_aligned, stress, sensitivity
fpe_industry_vertical	food_and_beverage, retail, ecommerce, hospitality, other
fpe_company_status	active, suspended, in_liquidation, archived

fpe_facility_status	available, partially_drawn, fully_drawn, repaid, cancelled, in_default

16. Entity Relationship Overview

The following table defines the primary foreign key relationships that connect the 10 schema layers. All relationships flow from the Planning Spine outward.

From Table (FK)	To Table (PK)	Cardinality	Relationship
scenarios	companies	Many-to-One	Each scenario belongs to one company
assumption_sets	scenarios	Many-to-One	Each assumption set belongs to one scenario
planning_periods	planning_calendars	Many-to-One	Periods belong to a calendar
plan_versions	scenarios + assumption_sets	Many-to-One	Version snapshots a scenario x assumption set
governance_events	scenarios / assumption_sets / plan_versions	Many-to-One (polymorphic)	Any governed entity can have events
markets	markets (self)	Many-to-One	Geographic hierarchy (parent_market_id)
kitchens	markets	Many-to-One	Each kitchen serves one micro-level market
kitchens	kitchen_formats	Many-to-One	Kitchen uses a format template
demand_drivers	scenarios + assumption_sets + periods + markets + platforms + product_families	Many-to-One (6 FKs)	Full planning grain key
price_plans	scenarios + assumption_sets + periods + markets + platforms + product_families	Many-to-One (6 FKs)	Full planning grain key
mix_plans	scenarios + assumption_sets + periods + markets + platforms + product_families	Many-to-One (6 FKs)	Full planning grain key
unit_cost_profiles	scenarios + assumption_sets + periods + markets + product_families	Many-to-One (5 FKs)	Cost grain key
labor_models	scenarios + assumption_sets + periods + markets + kitchens	Many-to-One (5 FKs)	Labor grain key
opex_plans	scenarios + assumption_sets + periods + markets + kitchens	Many-to-One (5 FKs)	OPEX grain key
capex_plans	scenarios + assumption_sets + periods + markets + kitchens	Many-to-One (5 FKs)	CAPEX grain key

equity_rounds	scenarios	Many-to-One	Equity round scoped to scenario
debt_repayment_schedules	debt_facilities	Many-to-One	Schedule belongs to facility
pnl_projections	scenarios + assumption_sets + periods	Many-to-One (3 FKs)	P&L grain key
cashflow_projections	scenarios + assumption_sets + periods	Many-to-One (3 FKs)	CF grain key
balance_sheet_projections	scenarios + assumption_sets + periods	Many-to-One (3 FKs)	BS grain key
unit_economics_projections	scenarios + assumption_sets + periods + markets + kitchens	Many-to-One (5 FKs)	UE grain key
kpi_projections	scenarios + assumption_sets + periods	Many-to-One (3+ FKs)	KPI grain key
scenario_comparison_snapshots	scenarios (x2)	Many-to-One (base + compare)	Compare two scenarios
driver_explainability_logs	scenarios + periods	Many-to-One	Explanation for a period/scenario
performance_alerts	scenarios + periods	Many-to-One	Alert in a scenario context
risk_scenarios	scenarios	Many-to-One	Risk scenario within a planning scenario
risk_registers	scenarios	Many-to-One	Risk register entry for a scenario
uncertainty_bands	scenarios + periods	Many-to-One	Band for a KPI in a scenario/period
stress_test_definitions	companies	Many-to-One	Company-level stress test library
risk_trigger_log	risk_registers + scenarios	Many-to-One	Trigger fires on a risk in a scenario
simulation_runs	scenarios	Many-to-One	Simulation based on a scenario
sensitivity_analyses	simulation_runs	Many-to-One	Sensitivity output from a run
monte_carlo_outputs	simulation_runs	Many-to-One	MC iteration output from a run
breakeven_analyses	scenarios + assumption_sets + periods + markets	Many-to-One (4 FKs)	Breakeven in scenario context
decision_memory	scenarios + markets + kitchens	Many-to-One	Decision recorded against context
portfolio_scenarios	companies	Many-to-One	Portfolio scenario at company level
market_portfolio_allocations	portfolio_scenarios + markets + periods	Many-to-One (3 FKs)	Allocation per market per period
cross_market_synergies	portfolio_scenarios + markets (x2)	Many-to-One (3 FKs)	Synergy between two markets

portfolio_capital_allocations	portfolio_scenarios + markets + periods	Many-to-One (3 FKs)	Capital per market per period
market_attractiveness_scores	markets + scenarios	Many-to-One (2 FKs)	Score a market in a scenario
revenue_projection_lines	scenarios + assumption_sets + periods + markets + platforms + product_families + revenue_streams	Many-to-One (7 FKs)	Full revenue projection grain
seasonality_profiles	companies + markets + product_families	Many-to-One	Seasonality per market/family
headcount_plans	scenarios + assumption_sets + periods + markets + kitchens	Many-to-One (5 FKs)	Headcount grain key
capital_deployment_plans	scenarios + assumption_sets + periods + funding sources	Many-to-One	Capital deployment schedule
debt_repayment_schedules	debt_facilities + periods	Many-to-One (2 FKs)	Repayment by period

17. Schema Summary — Table Count by Layer

Layer	Code	Name	Tables	Status
A	PSG	Planning Spine & Governance	8	Core v1
B	PMR	Product & Market Reference	8	Core v1
C	CDE	Commercial Driver Engine	5	Core v1
D	CTE	Cost Driver Engine	7	Core v1
E	FCS	Funding & Capital Strategy	5	Core v1
F	FPE	Financial Projection Engine	4	Core v1
G	KPI	KPI & Performance Intelligence	4	Core v1
H	RUF	Risk & Uncertainty Framework	5	Core v1
I	SDS	Simulation & Decision Support	5	Core v1
J	PCO	Portfolio Construction	4	Core v1
EXT	OEL	Optional Extension Layer	11	Phase 6+
		TOTAL	66	

Note: The total of 66 tables includes the 55 core tables across layers A-J and 11 optional extension tables. Core v1 delivery targets 55 tables across the 6-phase engineering roadmap (Phases 0-5). Optional extensions are delivered in Phase 6+.

18. Phased Delivery Alignment

The schema is delivered in phases aligned to the Engineering Roadmap v3.0. The table below maps each layer to its delivery phase.

Phase	Weeks	Deliverable	Layers	Tables Added	Cumulative Tables
0	1-4	Infrastructure & Schema Bootstrap	Setup only	0	0
1	5-12	Planning Spine MVP	A (full), B (partial)	8	8
2	13-24	Driver & Projection Engine	B (full), C, D (partial), F (partial)	+16	24
3	25-36	Scenario & Intelligence Layer	D (full), E, F (full), G	+19	43
4	37-48	Risk, Simulation & Capital	H, I, partial J	+10	53
5	49-60	Governance & Portfolio	J (full), governance enhancements	+2	55
6	61+	Operational Extensions	EXT (optional)	+11	66

19. Glossary

Term	Definition
AOV	Average Order Value — net revenue divided by total orders
Assumption Set	A versioned collection of planning assumptions attached to a scenario
CAPEX	Capital Expenditure — investment in physical assets (kitchens, equipment, technology)
COGS	Cost of Goods Sold — variable food and packaging costs per order
CLV	Customer Lifetime Value — total expected revenue from one customer over their lifetime
Contribution Margin	Net Revenue minus all variable costs (COGS + variable OPEX) per order
Dark Kitchen	A delivery-only food production facility with no dine-in capability
EBITDA	Earnings Before Interest, Tax, Depreciation, and Amortisation
EBIT	Earnings Before Interest and Tax (EBITDA minus D&A)
Finance-First	An architectural approach that prioritises financial projection accuracy over operational transaction granularity
FTE	Full-Time Equivalent — a unit of headcount measurement
IRR	Internal Rate of Return — the discount rate that makes NPV equal to zero
KPI	Key Performance Indicator
Macro	City-level geographic market unit
Micro	Area-level geographic market unit (neighbourhood, zone, GPS radius)
Monte Carlo	Probabilistic simulation that runs thousands of iterations with random inputs to model uncertainty
NPV	Net Present Value — present value of future cash flows minus initial investment
OPEX	Operating Expenditure — recurring operating costs (rent, utilities, labor, marketing)
Planning Grain	The dimensional granularity of a planning record (e.g., Kitchen x Platform x Product Family x Month)
Planning Grain Key	The composite of: company_id + scenario_id + assumption_set_id + planning_period_id
Planning Spine	The foundational entity hierarchy: Company -> Calendar -> Period -> Scenario -> Assumption Set -> Plan Version
ROMI	Return on Marketing Investment
ROIC	Return on Invested Capital
Scenario	A named version of the financial plan with distinct assumptions (Base, Bull, Bear, Stress Test)

Soft Delete	A logical deletion pattern using is_deleted=TRUE instead of physically removing records
Tornado Chart	A sensitivity chart showing drivers ranked by their impact magnitude on a target KPI
UUID	Universally Unique Identifier — 128-bit identifier used as primary key
WACC	Weighted Average Cost of Capital — blended cost of equity and debt financing
Working Capital	Current assets minus current liabilities; a measure of short-term liquidity

20. Document Control

Item	Detail
Document Title	FPE Data Model Schema v3.0 — Finance-First F&B Projection Engine
Version	3.0
Status	Approved — Master Reference
Date	2026
Architecture Reference	Unified Master v3 Architecture — Finance-First F&B Projection Engine
PRD Reference	FPE Product Requirements Document v3.0
Related Wireframes	FPE Dashboard Wireframes v4.0 (24 screens)
Engineering Roadmap	Phased Engineering Roadmap v3.0 (60 weeks, 7 phases)
API Specification	FPE API Specification v1.0 (Phase 1-2)
Scope	Food & Beverage — Dark Kitchen / Pizza Format — UAE Primary Market
Core Tables	55 core tables across 10 layers (A-J)
Optional Extension Tables	11 tables (Layer EXT, Phase 6+)
Total Tables	66 (55 core + 11 optional)
Total Enums	51 fpe_* PostgreSQL enum types
Primary Planning Grain	company_id + scenario_id + assumption_set_id + planning_period_id
Planning Horizon	6-18 months (core); 36 months (portfolio layer)
Currency Convention	DECIMAL(19,4); base currency AED; ISO 4217 currency_code
Database	PostgreSQL 15+ recommended; pgcrypto for gen_random_uuid()
Change History	v1.0 Initial schema (transaction-centric). v2.0 Finance-first rewrite. v3.0 Unified master — risk, simulation, portfolio, governance added. Scope fully aligned with Architecture v3 and PRD v3.